VANISHING OF THE HIGHER DIRECT IMAGES ALONG THE INCLUSION OF A QUASI-COMPACT OPEN IN A ONE-DIMENSIONAL AFFINE NOETHERIAN CASE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Commutation of higher direct images with flat base change. Stacks Project, Cohomology of Schemes, Lemma coherent-lemma-flat-base-change-cohomology (tag 02KH). Justifies reducing the stalkwise vanishing to the case of a local ring.
- A finite discrete scheme is affine; irreducible components of a Noetherian local ring. Stacks Project, Schemes, Lemma schemes-lemma-scheme-finite-discrete-affine (tag 0CKF); Algebra, Lemma algebra-lemma-Noetherian-irreducible-components (tag 00FZ). Supply the explicit description of U and its affineness in the local case.

Source reference. Original reference 02KH: sources/stacks-chow/tags.txt, source lines 3372–3372. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 00FR: sources/stacks-chow/tags.txt, source lines 581–581. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

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Source reference. Original reference 01XC: sources/stacks-chow/tags.txt, source lines 2513–2513. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let X be a scheme all of whose local rings are Noetherian of dimension ≤ 1. Let $U \subset X$ be a retrocompact open. Denote $j : U \to X$ the inclusion morphism. Then $R^p j_* \mathcal{F} = 0$, $p > 0$ for every quasi-coherent $\mathcal{O}_U$-module $\mathcal{F}$.*

Proof. We may check the vanishing of $R^p j_* \mathcal{F}$ at stalks. Formation of $R^q j_*$ commutes with flat base change, see Cohomology of Schemes, Lemma 02KH. Thus we may assume that X is the spectrum of a Noetherian local ring of dimension ≤ 1. In this case X has a closed point x and finitely many other points $x_1, \ldots, x_n$ which specialize to x but not each other (see Algebra, Lemma 00FR). If $x \in U$,

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then $U = X$ and the result is clear. If not, then $U = \{x_1, \ldots, x_r\}$ for some r after possibly renumbering the points. Then U is affine (Schemes, Lemma 02O0). Thus the result follows from Cohomology of Schemes, Lemma 01XC. $\square$